

Situated algorithms: a sociotechnical systemic approach to bias

Situated
algorithms

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Abstract

Purpose – The purpose of this paper is to propose that in order to tackle the question of bias in algorithms, a systemic, sociotechnical and holistic perspective is needed. With reference to the term “algorithmic culture,” the interconnectedness and mutual shaping of society and technology are postulated. A sociotechnical approach requires translational work between and across disciplines. This conceptual paper undertakes such translational work. It exemplifies how gender and diversity studies, by bringing in expertise on addressing bias and structural inequalities, provide a crucial source for analyzing and mitigating bias in algorithmic systems.

Design/methodology/approach – After introducing the sociotechnical context, an overview is provided regarding the contemporary discourse around bias in algorithms, debates around algorithmic culture, knowledge production and bias identification as well as common solutions. The key concepts of gender studies (situated knowledges and strong objectivity) and concrete examples of gender bias then serve as a backdrop for revisiting contemporary debates.

Findings – The key concepts reframe the discourse on bias and concepts such as algorithmic fairness and transparency by contextualizing and situating them. The paper includes specific suggestions for researchers and practitioners on how to account for social inequalities in the design of algorithmic systems.

Originality/value – A systemic, gender-informed approach for addressing the issue is provided, and a concrete, applicable methodology toward a situated understanding of algorithmic bias is laid out, providing an important contribution for an urgent multidisciplinary dialogue.

Keywords Gender studies, Bias, Situated knowledges, Algorithmic culture, Sociotechnical systems design, Strong objectivity

Paper type Conceptual paper

1. Introduction

In “Probabilistic Reasoning in Intelligent Systems” machine learning pioneer Judea Pearl (1988) pointed out that one of the big issues of “hard” logic of computing systems is its inability to address context-dependent information (p. 24). As computing systems are becoming ubiquitous and societies are being characterized as “algorithmic” rather than simply “digital,” the capacity to address context and constraints is becoming both a necessity and a challenge. This is especially salient, given the current state of debates around socio-cultural bias in algorithmic systems.

The burden of such bias is often borne by already marginalized populations. Recent publications show how algorithmic systems today, for instance, are in a position to deny health care with life-threatening, sometimes even deadly, consequences (Eubanks, 2018) and



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how they perpetuate racism and sexism through commonly used and trusted information providing platforms like search engines (Noble, 2018). Technological development is closely related to power dynamics and social, economic and political inequalities. The modern Western dichotomy between the social and the technical, however, makes it hard to address social issues such as bias in the development of algorithmic systems. Computer scientists and engineers oftentimes are not educated accordingly. Also, the complexity of the social world provides a challenge for any development process. Sociotechnical approaches (Bijker and Law, 1992) offer solutions by postulating the co-construction of society and technology. They require translational work that must be done between and across disciplines (cf. Lin, 2012).

This conceptual paper provides one example of such translational work. We propose that critical research and gender and diversity studies in particular, have a long track record of addressing bias and structural inequalities, thus providing a crucial source for analyzing and mitigating bias in algorithmic systems. We develop our line of argument along several key questions: what constitutes bias? How does bias operate? How does it form? Where to locate bias? How should accountability for bias be understood and allocated? What are the criteria for something to be considered biased and for whom? Is it possible to produce less biased knowledge through technological means?

The paper develops a new perspective and methodology for addressing bias in algorithms by bringing together feminist concepts of situated knowledges and standpoint theory with understanding of algorithms as sociotechnical systems. Through this, we provide a systemic approach to bias that responds to contemporary challenges of algorithmic culture.

2. Setting the scope: algorithmic culture and current perspectives on bias in information systems

2.1 Algorithmic culture and automated decision making

A sociotechnical perspective on algorithms asks how the social and the technical aspects are entangled in a co-constitutive manner. We understand algorithms simultaneously as computing technology and as formative for the social realm – a quality that we express through the term “algorithmic culture” (Striphas, 2015, p. 395). Ziewitz (2016) coined the expression of “algorithmic drama” (p. 3), stating that algorithms have almost become mythical objects: they are perceived as powerful agents but at the same time they seem to evade critical investigation. The view that machine learning is simultaneously pervasive and too complex to trace or even to understand grants algorithms power and agency to such an extent that it becomes almost impossible to oppose them or claim a critical take on their production.

To begin with, algorithms can be acknowledged as the “fundamental entity with which computer scientists operate” (Goldschlager and Lister in Goffey, 2008, p. 15). This fundamental role does not reveal any details of the connection between algorithms and society or culture, whereas the algorithm as the central means of computing is increasingly becoming a focal point for investigating social inequalities (e.g. O’Neil, 2016; Eubanks, 2018; Noble, 2018). In this way, we regard “algorithms [...] not [as] autonomous technical objects, but complex sociotechnical systems” (Sievers, 2018, p. 378). A situated account then delivers the insights and tools to reconstruct and understand algorithms as sociotechnical systems. But what exactly does such an understanding imply?

A sociotechnical framing of algorithms avoids a simplified perspective that opposes algorithms and humans and situates algorithms and their effects within “our” human living world. To understand how algorithms, society and culture are entangled, we identify two overlapping, but differentiable, levels of entanglement: first, the level of social inequalities that are reproduced through conjoining a practical purpose and automated decision making; second, the level of cultural work of algorithms in sorting, classifying and hierarchizing, which Striphas (2015) calls an “offloading of cultural work onto [...] digital technologies” (p. 395), forming an “algorithmic culture.” At its core, algorithmic culture analyzes “the

enfold of human thought, conduct, organization and expression into the logic of Big Data and large-scale computation” (p. 396). Hence, the algorithm becomes a centralized form or structure that shapes human existence in a grounding manner, functioning as the “key of modern rationality” (Totaro and Ninno, 2014, p. 29).

In short, “[a]lgorithms do things, and their syntax embodies a command structure to enable this to happen” (Goffey, 2008, p. 17). They also need a physical machine to do things, such as a personal computer. Furthermore, what algorithms do can be divided into two processes: first, the process of automating decision making by transforming an input into an output along a defined set of steps; second, the process of engineering that also includes the “pragmatic efficacy of building software for particular purposes” (Goffey, 2008, p. 15). In this regard, a practical purpose and a set of mathematical steps are conjoined, forming a specific algorithm, for example one of the algorithms constituting a search engine. Notably, this practical purpose emerges from the contexts in which an algorithm is programmed, tested and improved – a process not possible without humans and their expertise knowledge in whatever field might be relevant for that specific algorithm.

For example, search engines become political matters as they are “determining any systematic inclusions and exclusions [as well as] the wide-ranging factors that dictate systematic prominence for some sites, dictating systematic invisibility for others” (Introna and Nissenbaum, 2000, p. 171). Inclusion/exclusion, and invisibility/visibility depend upon how purpose and decision making are joined in the algorithms of search engines through practices and context of the algorithms and their construction. Other algorithms, like those for music recommendation or those used for people identification, also demonstrate the co-shaping of social and technological structures (cf. Sievers, 2018).

2.2 Knowledge production and bias identification in algorithmic systems

The two ways, outlined above, in which algorithms operate as sociotechnical systems have lately become more transparent through works of scholars such as O’Neil (2016). She not only problematizes the role of computer-aided decision making but also highlights the importance of choices made by developers: “The math-powered applications powering the data economy were based on choices made by fallible human beings” (p. 3). While these choices become “opaque,” the effects of the “encoded human prejudice, misunderstanding, and bias into the software systems” (ibid.) result in an increase of inequality. Here the effect of producing inequalities is grounded in the mathematical models themselves, making it critically important to investigate the biases as well as the alleged neutrality and objectivity of mathematical models. This perspective brings together questions of social inequality and the cultural work of designing algorithms and collecting data. Viewing algorithms as sociotechnical systems also implies a broadened understanding of bias in information systems.

Throughout scientific literature, bias is often identified to come from one of the three sources: bias in data, bias in algorithms themselves (including technical constraints and considerations) and bias in the context of use or “emergent bias” (Friedman and Nissenbaum, 1996)[1]. The first category is also known as “garbage-in, garbage-out” model, and it is based on the idea that the results can be skewed because data and the society, from which they are derived, are biased. Such “pre-existing bias” entails broader biases that have their “roots in social institutions, practices, and attitudes” (Friedman and Nissenbaum, 1996, p. 333), including decisions about which data are collected and by whom, for which purposes, how they have been processed, etc. Technical bias[2] – bias in algorithms – arises in the technical features of the system, such as inaccurate application of a model or over-reliance on formalized human constructs. Technical bias can include certain mathematical tendencies, such as reliance on homophily (Chun, 2016) or tendency to minimize differences through normalization (McQuillan, 2018). Lastly, “emergent bias”

relates to the use of the system as a whole and arises from the mismatch between design assumptions and use patterns.

Nissenbaum and Friedman provided a working definition of biased systems, as “computer systems that *systematically* and *unfairly discriminate* against certain individuals or groups of individuals in favor of others” (p. 332, emphasis original). This raises questions over definitions of fairness and less-biased technological systems design. Reporting on computational methods that scientists are developing to counter algorithmic biases, Courtland pointed out that the question of fairness of algorithms is not only a philosophical but also a technical one: “[p]redictive parity, equal false-positive error rates, and equal false-negative error rates are all ways of being ‘fair’, but are statistically impossible to reconcile if there are differences across two groups” (Courtland, 2018, n.p.). Narayanan (2018) highlighted the different definitions of fairness and argued for a need for governments (and, we would add, all organizations relying on algorithmic systems) to deepen their understanding of what such systems can do, what the definition of optimal performance is and how to set the parameters for such performance.

The research scope of examining bias was even wider for other researchers. Crawford (2017, 2018), for instance, repeatedly pointed out that it is not only the biases within (historical) data, but also the very modes of algorithmic knowledge production through categorization and classification that we need to interrogate to understand better their workings as well as the resulting biases. Crawford argued, similar to Bowker and Star (2000), that classification and categorization are essential for human scientific knowledge production and structural organization of society. This also has inherent limitations – historically, classification systems have been used to advance imperialist, colonialist and racist agendas (e.g. Mahmoud, 1999; Schiebinger and Swan, 2007). Hence, it is crucial to investigate how we categorize, what we intend to classify and for which purposes, how we perform the aforementioned offloading of cultural labor onto technical systems – both on the level of data (how data are collected, labeled and which categories are used) and on the broader level of the purpose and performance of an algorithmic system.

Finally, language and discursive setting should also be considered. As noted by Agre (1997), technical field concepts – like “information,” “intelligence” and “knowledge” – are used both in mathematical, formalised terms, as well as in colloquial terms. According to him, such blurring of technical and colloquial definitions points to the discursive setting of a technical practice that both enables fruitful crossovers between disciplines but also naturalizes certain concepts and their meanings and allows us to take them for granted. Bias in information systems similarly has a double connotation. It has been defined in technical terms, but it is also important to interrogate how the colloquial, everyday notions of bias play a role in systems development and how these “technical” and “vernacular” definitions coalesce in practice.

2.3 Current debates and approaches to moderating social and cultural bias

The entanglement between algorithms, society and culture becomes even more prevalent in debates around addressing and “solving” bias in algorithms. The increased awareness of social and cultural bias in algorithms negatively affecting marginalized people gave rise to a series of proposals for countering or mitigating adversarial societal effects. Awareness-raising publications and guidelines (Diakopoulos *et al.*, 2017; ACM US Public Policy Council, 2017), calls for algorithmic and data transparency[3] as well as contributions to discourses around accountability and practical approaches for tackling bias in algorithms are on the rise. In the following, we provide an overview of major discussion threads.

Algorithms that are at the center of attention when it comes to the (re)production of social inequality are data driven and mainly belong to the class of the so-called artificial neural networks[4] and deep learning. A common metaphor for the math behind these algorithms is

that of a learning entity, which from a knowledge base (the training data) tries to generalize in order to cope with unknown situations. Just as the quality of the practical purpose, most training data are created and curated by people and thus necessarily reflect parts of their knowledge, assumptions and values and their socio-cultural-political context in general. One straightforward approach to making data uncompromisingly transparent is raised in political discussions around constraining private ownership of data set(s) and instead collectivizing them (Srnicek, 2017, p. 68). Nevertheless, due to the vast complexity of training data sets[5], meaningful insight is difficult to achieve. One proposal to provide means to “efficiently compare, select, and interrogate datasets” was developed by Holland *et al.* (2018): their “dataset nutrition” (<http://datanutrition.media.mit.edu/>) framework allows inquirers a general overview regarding data set’s variables, metadata, provenance and basic statistics, and also offers modules to investigate linear correlations. However, markers of social inequality are often redundantly encoded in (multiple) non-obvious proxy variables that correlate strongly with belonging to a protected social category (gender, race, class, etc.) (Dwork *et al.*, 2011; Barocas and Selbst, 2016). For example, knowing where someone lives may serve as an indication of their ethnic background without violating anti-discriminatory imperative of omitting such categories from algorithmic decision systems.

Regarding algorithms, Seymour (2018) argued that due to the internal complexity of the current deep learning algorithms, the focus should be on outcome transparency instead of process transparency. In this context, it is fundamental to realize the difference between relatively static statistical predictor or scoring algorithms that work with a limited set of data and a given (array of) formulas, as discussed by O’Neil (2016), and (assemblages of) deep learning algorithms that access lots of different types of data and internally create their own concepts and correlations “in-process.” Wang and Kosinski (2018) reported to have created a neural network that is able to identify one’s sexual orientation from a picture of their face “better than humans”; Christl (2014) reported on algorithms from different companies that create and combine profiles of people’s psychological, physical and social situation based on their behavior on the internet. In such cases, it is uncertain which variable(s), features or internal algorithmic processes lead to these results. Thus, even if such complex algorithms would be made transparent, very few people or auditors, if any, would be able to understand an algorithmic decision, since often not even their creators are able to understand their internal processes.

Several researchers try to make the inner workings of some types of machine learning processes more comprehensible. Termed “explainable AI” (Samek *et al.*, 2017), the idea is to support human understanding of individual results as well as what a given model has learned in terms of internal representation. Exemplarily, there are several techniques for understanding deep learning models for image classification tasks, such as feature visualization[6] (Olah *et al.*, 2017). Another example is an approach to automatically generate counterfactual explanations (Wachter *et al.*, 2017), where (negative) decisions made by algorithms are supplied with information about which part of someone’s input data is responsible for the decision in question, thus providing indications of what needs to be changed in order to get to a desirable outcome.

Practical approaches to avoiding instead of explaining unjust results of algorithmic decisions have also been developed. Bolukbasi *et al.* (2016) offered a methodology to remove gender stereotypes from word embeddings – a natural language processing technique that reduces the mathematical dimensionality of words’ or phrases’ representations and thus makes them easier to process computationally. Fairness criteria like demographic parity (Zemel, 2013) and equal opportunity (Hardt *et al.*, 2016) prohibit discrimination against protected attributes on a statistical level in the internal model of an algorithm – but may be useless, as argued above, if protected attributes are encoded redundantly in proxy variables. Furthermore, the selection of criteria is often mutually exclusive and depends on the preferred definition of

fairness and context. For example, the equal opportunity criterion has been specifically designed to optimize the ratio between fairness and profit – to minimize the cost of fairness.

The first steps toward public oversight, understanding and empowerment have also been reported. Buolamwini and Gebru (2018) called for a close examination of who, how and why people produce code and founded a participatory and interdisciplinary initiative Algorithmic Justice League (www.ajlunited.org/) to provide a platform for reporting as well as identifying algorithmic bias, promoting inclusivity in training sets and raising consciousness for sociotechnical aspects of contemporary algorithms. Additionally, recently published algorithm bias audit open-source toolkits like Aequitas (<https://dsapp.uchicago.edu/projects/aequitas/>) and AI Fairness 360 (Bellamy *et al.*, 2018) aim at facilitating the examination and reporting of potentially existing bias in an algorithmic decision by evaluating open data sets utilizing, among others, the aforementioned statistical fairness criteria.

Although (semi) automated algorithmic decisions might have a devastating impact on the most vulnerable societal groups (Eubanks, 2018), how to ensure algorithmic accountability for the algorithm's adversarial effects is still unclear. As pointed out by Kroll *et al.* (2016), "the accountability mechanisms and legal standards that govern decision processes have not kept pace with technology" (p. 636). They proposed two sets of approaches: the first one aims at ensuring the expected functioning of an automated decision from the start of the design process, and the second allows review and oversight while the algorithm is in place. Interestingly, they solely put their trust in "technical tools" for ensuring accountability, arguing that "[...] computerized processes can be designed for governance and accountability" (2016, p. 640). In the following parts, we argue that there can be no such thing as a "purely" technical tool, but rather an assemblage of sociotechnical entanglements, and we provide arguments for the need of a multidisciplinary, gender-informed and systemic approach to tackle bias in algorithms.

3. Reframing the question of bias in information systems through an intersectional gender studies perspective

3.1 Situatedness, "strong objectivity" and rethinking systems of classification

Gender studies is a diverse field that deals not only with relations among genders but critically reflects on systems of classification as such (man/woman, nature/culture and others) and asks how these systems re/produce social inequalities. Inequalities are indexed on social categories (race, class, gender, ethnicity, sexuality and dis/ability) that can work simultaneously and form complex patterns of power and hierarchy – Crenshaw (1989) called this "intersectionality." Feminist epistemology shows that these patterns in turn influence what counts as valid knowledge and as objective fact, whereas the binary, hierarchized view of gender relations has historically served as a ready-made model for interpretation of scientific findings[7].

Gender studies has a long track record of interrogating structural biases that permeate social relations and political structures, as well as scientific knowledge production and technology design. An example of structural bias has been documented in evolutionary biology, where the paradigm of the "survival of the fittest" and the resulting explanation of evolution as a competitive struggle were shown to be masculinist, obscuring the role of collaboration and symbiosis in evolution (Margulis, 1991, 1998). As for technology design, gender often served as a lens for what counted as "technology proper"; for instance, software programming was first considered a clerical, administrative task performed by women. However, when it became acknowledged as a technological field, it was rapidly "masculinised" on a symbolic and practical level, and as a result, women were pushed out of it (Ensmenger, 2010). The perspectives of gender studies also show that the lack of involvement of women and other marginalized groups in technology design can result in decreased usability; for example, Steinmann *et al.* (2013) showed how the lack of women mappers in open-source mapping projects generates biased maps, whereas Boyd (2001)

interrogated how certain technical aspects of 3D virtual reality design resulted in these systems being less usable to women.

Due to focus on power and structural inequalities, gender studies hold potential for productive investigation of bias in information systems. For this, we elaborate on two concepts in particular: “situated knowledges” (Haraway, 1988) and “standpoint theory” with its related concept of “strong objectivity” (Harding, 1991, 2005, 2015). Situated knowledges theory draws attention to the fact that knowledge production cannot be cut off from the positionality of researcher and context. According to Haraway, the idea that scientists are neutral observers, detached from socio-political context, emotion, biological ties and any other kind of positionality, is a masculinist ideal of the “view from nowhere” that obscures the fact that all knowledge is produced by specific bodies, in specific material and cultural conditions. If science is supposed to “see the truth,” then we need to keep in mind that the “seeing” itself and organs of vision (be they the eyes or the many tools that are used for scientific research) are embodied (biologically, technically and culturally). Our forms of vision allow us to see certain things and at the same time obscure others, while where we see from implicates us in the web of power relations and influences both how and what we can see. This partial perspective, if not reflected upon, becomes a problem in sociotechnical systems design whenever the developers take their own perspective as representative of the end user[8]. Thus, we need to re-think scientific knowledge production not as universal but rather as valid from a specific perspective or position that operates always within certain figurations of time, space and artefacts, which is to say: as situated knowledges[9].

Such a position resonates well with some of the contemporary conceptualisations of bias in information systems, for instance, as proposed by Schiebinger and Zou (2018). According to them, to counter bias effects of algorithms, we need to diversify data in terms of geodiversity and demographics as well as pay attention to the imbalances in power relations. We argue that situated knowledges perspective provides a methodological and systemic approach to how these power imbalances and diverse positions can be accounted for. Furthermore, the benefits of situatedness are not a supposed neutrality or unbiased nature of knowledge, but rather the possibility to account for the knowledge produced through the sociotechnical system in question.

Harding’s standpoint theory and her related concept of “strong objectivity” specifically address the question of power differences and resulting differences of benefit of scientific knowledge and technological systems. According to Harding, those involved in knowledge production must ask themselves the following questions: whose perspective are we looking from? Who benefits from this perspective? Who is at a (structural) disadvantage? Scientific knowledge production is always already embedded in a broader socio-political context; thus, it cannot escape power relations. This becomes clearer when we look at historical cases where science (anthropology, as well as medicine and biology) was used to discriminate against black populations and provide grounds for the theory of racial supremacy. Technologies, therefore, are also part of the broader set of power relations, whether their creators are conscious of it or not. For instance, one could argue that facial recognition software’s failure to recognize faces of color further extends the racial inequalities present in training data sets, which, in turn, reflect the unequal representation of race in general.

In response to such existing webs of unequal power dynamics that influence knowledge production, Harding argued for a “strong objectivity,” which is generated when knowledge is produced from the perspective of those, whom it affects the most. For her, pure, “value-neutral” objectivity (similar to Haraway’s “view from nowhere”) is in fact “weaker” because it does not take into account its own positionality and entanglements in power relations. “Strong objectivity” is, thus, enabled in those research endeavors that start from interrogating the existing inequalities, reflect their own position in the system of these inequalities and actively ask which constituencies will or will not benefit. For example, if we are to explore the structure of racial inequalities in the Global North, a stronger, more

accurate account of such structures would be produced by people of color or ethnically diverse population rather than by white and ethnically homogeneous population.

Harding's standpoint theory resonates with contemporary perspectives in fields such as human-computer interaction for development and information communication technologies for development. These have been criticized previously for exporting western, Eurocentric universalist perspectives to the Global South and so-called "developing countries" (e.g. Merritt and Bardzell, 2011; Leye, 2009). More recently though, postcolonial and decolonial computing perspectives as well as participatory design methods highlighted the need, as argued by Harding, to not only pay attention to the perspectives of marginalized/oppressed groups, but also to directly include them in the process of knowledge production and technology design (Irani *et al.*, 2010; Philip *et al.*, 2012; Ali, 2016). Furthermore, by focusing on the question of who benefits from and who bears the losses for a particular technological solution or scientific research, standpoint theory raises the question not simply of fairness but particularly that of justice as well. This line in practice is taken up also by Buolamwini in her work with Algorithmic Justice League and her research on facial recognition systems (Buolamwini and Gebru, 2018) and by the Data Justice Lab team (<https://datajusticelab.org>).

3.2 *Understanding and defining bias through the lens of gender studies*

The understanding of what constitutes bias from a perspective of gender studies builds on feminist epistemology. Against this background, there is no unbiased knowledge. All knowledge has its roots in a specific time, space and context. What counts as knowledge and whose knowledge counts or is accounted for are intertwined with socio-political power relations and subject to historical change. Gender studies respond to this through various methodological reconsiderations. Feminist approach to empiricism (experience-based knowledge production), for example, seeks to collect the experience of marginalized or overlooked positions (e.g. women and people of color) to achieve broader empirical accuracy and contribute to "strong objectivity" (Harding, 2005). This also fosters participation in and access to scientific knowledge. By doing so, feminist empiricism implements a specific value of raising awareness of gender bias while also giving more just accounts of the world in its research design. The goal is "to reveal both gender in the phenomena and gender bias in the accounting of them" (Longino, 1997, p. 45). Feminist scholars of science have also shown how gender bias has tainted the presumed objectivity of scientific knowledge and hindered scientific innovation and progress (cp. Lloyd, 2006; Fausto-Sterling, 1985). Among other aspects, gender bias affects the choice of topics for research as well as research design and methodologies. In medicine and public health, suggestions for taxonomies to identify and prevent bias have been made that take up feminist history of science research as a basis. They reach back to early works, such as non-sexist research methods and gender-based analysis (Eichler, 1988, 2001a). Follow-up works include a framework for analyzing gender bias in public health and epidemiology (Ruiz-Cantero *et al.*, 2007). Common to these approaches is the claim that lack of gender awareness in concept or methodology causes systemic gender-dependent errors and has profound effect on research outcomes and on women's health.

For a systemic approach of tackling bias in algorithmic decision-making systems, we take the aforementioned gender research into account and also various forms of gender bias outlined below. Being able to identify gender bias is the first step toward raising awareness and opening debates about sociotechnical countermeasures. In gender-based analysis, Eichler (2001b) named three superordinate problems of gender bias: androcentricity or androcentrism, gender insensitivity and double standards (p. 18). Each superordinate problem has certain sub-forms, characteristics and manifestations.

Androcentrism describes the focus on the masculine or male perspective. This, as with many biases, is an unconscious move, very common and hence very potent in its constitutive power. The perspectives of men – typically white, able-bodied and heterosexual

– are taken as normative and representative, whereas perspectives of women, gender non-conforming people or men that do not fit the norm are marginalized or excluded. With the lack of diversity in technological fields, androcentrism is prevalent and has effects on work culture and technological progress (DuBow, 2013; Schiebinger *et al.*, 2011–2018). In engineering, androcentrism manifests in standards that have historically been modeled after able-bodied men (e.g. crash test dummies, electronic speaking aids and prosthetics) and that had to be diversified in order to include children and women. In computing, androcentrism, together with “I-methodology” (Rommès, 2002) and lack of ethnic, gender and racial diversity, has led to failed, un-usable for some or not-accepted technology (e.g. facial recognition that worked best for white men and worst for black women, as pointed out by Buolamwini[10]). When people of specific gender or ethnicity are excluded from data sets that inform research and this is not reflected upon, it tends to produce overgeneralization: a certain percentage becomes normative for all others. Due to data availability and the way most data sets for machine learning algorithms are collected, overgeneralization is a problem for AI systems. Schiebinger and Zou (2018, p. 326) referred to a study of the ImageNet data set as training data for image recognition, where more than 45 percent of data come from the USA, which makes up only 4 percent of the world population. China and India combined bring only 3 percent of data, even though they make up 36 percent of the world’s population. Feminist empiricism makes a strong point in countering androcentrism by enriching data sets. Another sub-form of androcentrism is paradox gynocentrism, meaning that in fields that are considered typically female, men are excluded from the data set.

This provides a connecting point to the second superordinate problem: gender insensitivity. A major challenge of gender-informed research is assessing what role gender and intersecting categories play in a research field, not to overestimate or underestimate the relevance of gender. Insensitivity toward gender or ethnicity affects research through decontextualization, which leads to ignoring that different genders are differently affected by societal structures. Situations or places that seem to be the same for everyone can be very differently experienced, depending on gender, religious background, sexual orientation or ethnicity (cp. Lundman, Kaufman, 2003)[11]. Familialism or householdism bias also relates to insensitivity: if household data are collected, it is important what is considered a household (nuclear heterosexual family, single-parent households, same sex couples, shared and intergenerational living, etc.) and how gender relations affect the household. If hegemonic normative views on family structure are employed, the collected data fail to represent the complex reality of family life. In consequence, this insensitivity may lead to algorithmic systems unknowingly incorporating familialism bias.

The third problem – double standards – means assessing the same situation, context, behavior or characteristics differently based on a person’s gender. Given the structuring power of the gender order, examples for overt and hidden double standards influence everyday life as well as research practices. A common mistake leading to gender bias in falsifying data collection and research results is related to double standards that inform the early stages of research. Eichler (2001b, p. 18) provided an example from public health: a higher number of poor health symptoms of women were attributed to their hormonal cycles, when actually their working conditions were to blame. The example links to the sub-form of stereotyping gender traits: ascribing certain values/expectations/character to different genders, often reproducing prejudices that are well established (here women as ruled by emotions and hormonal imbalance). Another sub-form is emphasizing gender dichotomies, which entails focusing on showcasing differences between genders where such differences are not majorly significant.

We end this section with a disclaimer: the discussed criteria refer to gender bias. They do not explain racial bias or class discrimination, or how various categories intersect with each

other. Androcentrism, in particular, takes up white men as the norm, as noted above. Furthermore, the category of “gender” itself is being questioned in transgender scholarship and black feminist thought. Thus, for future work, gender bias criteria must be broadened through intersectionality. As pointed out in Section 3.1, there are excellent studies on the relation of class and race (e.g. Hooks, 2000; Race and Class, 2018) that can expand the stereotyping bias beyond gender.

4. Suggestions for less-biased information systems: toward a systemic, situated sociotechnical approach

Two gender studies perspectives – situated knowledges and standpoint theory – were introduced above, in addition to how gender bias is conceptualized in gender studies. In this section, we bring these concepts together and argue that situating algorithmic systems and embracing standpoint theory generate a more systemic understanding of bias and accountability. We see our approach as congruent with other critical and design-oriented solutions to mitigate bias that are underway in data science and computer science. What we contribute is a conceptual framework that is distinctively rooted in feminist theories, attuned to inequalities and power dynamics and that takes sociotechnical nature of information systems as a starting point. Specifically, we suggest that the perspectives of situated knowledges and standpoint theory point to understanding knowledge as a product of a complex network, where human researchers, data, data structures, algorithms and broader social, political, historical and scientific contexts all contribute to the specific results that are produced (cf. Akrich, 1992). This, in turn, invites rethinking bias as a complex phenomenon, distributed across the whole process of designing information systems.

4.1 Diversification of data sets/sources

Similar to many other scholars, we agree that there is a strong need to (geo)diversify (Schiebinger and Zou, 2018) data bases and sources used for algorithmic prediction and classification, particularly for training the algorithms to do such work. However, we suggest two additions: first, system designers should not only ask whose data are being used, with assurance of a balanced “dataset nutrition” value (Holland *et al.*, 2018), but they should also ask which types of data are needed for the system at hand. For instance, although many data sets by default include information about gender, it should be asked whether gender as a category is indispensable for the system that is being designed or whether it leads to emphasizing gender dichotomies. Second, in making sure that data sets include a diverse pool of people, attention should be paid to historical power imbalances when collecting the missing data. If the data are collected about already marginalized populations, it should be considered whether these populations will then benefit from inclusion into data sets or whether they will be further stigmatized and harmed[12]. An example is the disproportionately high surveillance and policing of populations of color across many Global North countries. Therefore, it should be asked: “Do the marginalized populations want to be represented, which could potentially further increase surveillance?”[13]. The approach of feminist empiricism can be particularly helpful for practices of data collecting and classifying that are more inclusive, aware of social inequalities and the effects of classifications. For certain groups, the claim for participating power could also entail the right to refuse being a source for information.

4.2 Diversification of approaches to design

Following the ethos of “strong objectivity,” we suggest that it is beneficial to map out which segments of a community or which populations will be affected by the system or new technology and include (representatives of) these populations in the design process from the

start. In this way, the potential biases that seep into the research and design framework (such as the aforementioned androcentrism or paradox gynocentrism, overgeneralization and insensitivity to gender) could be prevented. Computer science has many socially responsible or human-centered design approaches readily available, including value-sensitive design, user-centered design, inclusive design, participatory design. These approaches are located in human–computer interaction. For machine learning-based information systems, they must be revisited and updated (Alvarado and Waern, 2018). AI brings new challenges to the distribution of agency between humans and machines and interaction design: from the initial planning stages to the usability of the interface, questions of participation, as well as principles like transparency or accountability, need further research. Furthermore, although proprietary software and algorithms are sometimes hard to bypass or avoid, it should be investigated whether more open source, community-built tools can be employed for design and implementation of systems, to increase both the sense of empowerment of the stakeholders and transparency of systems themselves. This is especially salient for lower scale, community-oriented systems.

4.3 *Situating: including contextual factors in a systemic way*

To produce less biased and more accountable sociotechnical solutions, it is crucial to situate algorithmic systems and their design process, i.e. to understand and address their embeddedness in political, socio-cultural contexts and existing power structures. This situatedness should manifest in all steps of the development process, from defining the problem or issue that the system is supposed to solve or improve on, setting the goal/function of the system, to selecting data sets and classifiers, to deciding upon the training process, to running tests and to application “in the wild.” This could be achieved by systematically and iteratively addressing the following “4P” set of questions:

- *People*: who are the people affected and involved? Who is benefiting? This helps to avoid the abovementioned decontextualization and encourages the resulting design framework toward strong objectivity by paying attention not only to who is interested in developing the system (who benefits) but also which other constituencies are affected in more negative ways. By laying out the map of involved stakeholders and pointing out the differential effects of the system, its accountability is also encouraged. To take the example of one of the biases outlined in Section 3.2, this line of questioning helps avoid decontextualization and gender insensitivity.
- *Place*: where are the data coming from? What is contained in data and how they have been collected? How will the system affect the sources of data, if at all? This latter question is especially important because it encourages to investigate potential consequences, especially nowadays when it has been proven that if there are structural inequalities embedded in the data, they are likely to be exacerbated via algorithmic systems (O’Neil, 2016).
- *Power*: what are power hierarchies between the initiating parties, the benefiting parties and others that will be affected by the algorithmic system? Should/could these power relations be made more equal? This concerns not only the evaluation of the system once it is already designed, but also requires a critical reflection on the data as well as design process. The considerations regarding power span across time and include historical trajectories of power relations as well as how these relations are solidified into structural inequalities. Such historical and structural approach also includes looking at intersecting inequalities (gender–race, race–class, gender–class, etc.) and might help prevent encoding by proxy.

- *Participation*: who participates in the design? What kind of technological, social and cultural systems will play a role in the application of the system? Participation here is to be investigated both in terms of human participation and involvement (stakeholders and group diversity), as well as non-human participation (methodologies, tools, techniques and modalities). Considering the other systems to which the particular algorithmic system would relate to in practice, as well as taking into account potential outcomes, is also crucial for grounding and situating.

Answering these questions requires diverse disciplinary expertise and methods. Echoing Schiebinger and Zou (2018), we argue that interdisciplinary teams and work culture are needed.

4.4 Setting due process: addressing fairness, transparency, justice and accountability

We suggest that the debates around accountability, transparency and fairness of algorithms should include investigation of the definitions of these terms within the context of groups that will be most affected by the prospective system in question. Asking whether the system would increase/decrease the agency of any of the affected groups should be added to the list of considerations. In addition, although the values of fairness and transparency are usually addressed, justice, including social and environmental justice, can also be added to this list – especially when the systems in question are to be employed in the public sector or by governmental institutions.

Considerations of justice resonate with already existing debates on accountability. Following strong objectivity and standpoint theory, we argue that in order to develop better accountability standards and reduce biases occurring through the development process, the purpose and the expected results of a particular information system should be better examined. For instance, a closer look could be paid by designers and researchers to the concepts that are used to describe what information systems do and how the meaning of those concepts shifts in different contexts. For example, the notion of “prediction” for developers might mean that the system analyses data and discovers patterns that express strong correlation. However, once such a system is used for the purposes of drawing policy suggestions (e.g. in “predictive policing”), “prediction” acquires a different, more colloquial meaning, affecting the expectations of the user and the (mis)understandings regarding the kind of knowledge generated by the system.

Lastly, when it comes to accountability and responsibility, responsibility in computerized society needs to be re-defined in a more dispersed and nuanced way since the ownership of blame does not follow a clear, linear path but is instead more scattered through a network of actors (Nissenbaum, 1997). Thus, a more “distributed” understanding of bias as occurring throughout the process as well as a distributed approach to accountability, and perhaps indeed also to fairness and justice (to whom is the system fair or just? How is fairness and justice distributed?) should be developed.

5. Conclusion

In this paper, we introduced a sociotechnical approach to bias in information systems. Specifically, we drew on gender studies’ rich expertise in addressing issues of inequalities, discrimination and inclusion/exclusion. Furthermore, on an epistemological level, gender studies and feminist theory question the modern paradigm of scientific objectivity as impersonal, a-political and therefore normatively neutral. In contrast, feminist epistemology, science studies and feminist empiricism focus on *whose* knowledge is accounted for in science, *where* this knowledge comes from and *what forms* of experience or knowledge can be expressed as data or even broader through computational means. Perspectives like these seem to be largely absent from the current development of algorithmic systems.

We demonstrated this by revisiting definitions of socio-cultural bias in information systems and some of the possible solutions that are currently discussed. This revisiting backed up our finding that the inability to contextualize and situate technological development properly within the world and lack of sensitivity to societal power relations yield problematic outcomes.

Acknowledging the fact that computing and algorithmic systems exist in a broad set of sociotechnical conditions and analyzing the co-construction of algorithms, society and culture is only one part of the story. We are aware that accounting for the complexity of socio-cultural, political and economic dimensions while developing information systems is even more challenging.

As we have pointed out in the previous section, bridging the prevalent divide between the social and the technical toward a systemic sociotechnical approach first and foremost calls for true interdisciplinarity. Translational work in teams needs experts from application domains, but perspectives of end users and people possibly affected by the developed systems should also be included. There is also an increasing need for interdisciplinary methodologies, methods and tools connecting critical knowledge from the humanities and social sciences to computing. Our conceptual paper provides one example of such translational work. We are aware that our suggestions for less-biased information systems are not easily or readily implementable. Instead, they provide a set of guidelines and are meant to help navigating the complexity of developing less discriminatory sociotechnical designs. Future work would, thus, include polishing the suggestions further, developing a more applicable framework or a process model and, of course, testing and evaluating their validity.

Notes

1. More specific lists of bias sources can be identified for various systems (e.g. see Baeza-Yates (2018) for bias in web systems).
2. “Technical bias” is originally used by Friedman and Nissenbaum. They state that part of it is the over-reliance on formalized social constructs, thus even on the so-called technical level bias is always already sociotechnical.
3. The discourse around transparency of algorithms and data focuses mainly on the means of understanding the results and auditability of algorithmic processes, which is, thus, conceived as beneficial for (affected) individuals and society. For a more cross-systemic view of transparency and accountability, see Ananny and Crawford (2018).
4. Artificial (deep) neural networks constitute the fundamental framework for the most successful machine learning algorithms today. Such network is made of layers of “neurons” or nodes that propagate (and potentially alter) the incoming signal from one layer to the next until it reaches the end and provides a result. The basic idea is to provide the network with lots of examples, e.g. images of human faces, and the desired result (“human face”), and let it build up its own internal representation by altering the weights between the layer’s nodes until it can detect faces in unknown images (the test or real-world data). For a comprehensive overview, see Schmidhuber (2015).
5. For example, the (open access) ImageNet data set (Russakovsky *et al.*, 2015), commonly used to test the performance of image classification algorithms, contains over 14m labeled images.
6. Feature visualization tries to visualize how neural networks for image classification tasks develop their internal representation. See <https://distill.pub/2017/feature-visualization/>.
7. E.g. Keller (1985, 1995, 2002) presented a critique on how the gendered view influences the way biological findings are interpreted and presented.
8. Rommes (2002) referred to this as “I-methodology.”

9. The contingency of knowledge is also highlighted in the findings of quantum physicists, especially the experiments of Bohr, through which he proved that depending on the instrument of measurement, light can act both as a particle and as a wave. For more on Bohr's research and how quantum physics intersects with feminist theory and epistemology, see Barad, 2007.
10. "Facial Recognition Is Accurate, if You're a White Guy", *New York Times*, February 9, 2018, www.nytimes.com/2018/02/09/technology/facial-recognition-race-artificial-intelligence.html
11. For example, spaces that employ facial recognition or body scanners, such as airports, might be experienced as problematic or threatening by transgender people or refugees (Currah and Mulqueen, 2011; Latonero and Kift, 2018).
12. Further proposals and approaches to data science from a feminist perspective are being developed in the forthcoming book by D'Ignazio and Klein (forthcoming). A draft is currently published for community review at <https://bookbook.pubpub.org/data-feminism>.
13. For further references on this debate, see Flyverbom (2016) and Harwell (2018).

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